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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 28: TOD Case Study-III Naya Raipur

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Dept. of Civil Engineering



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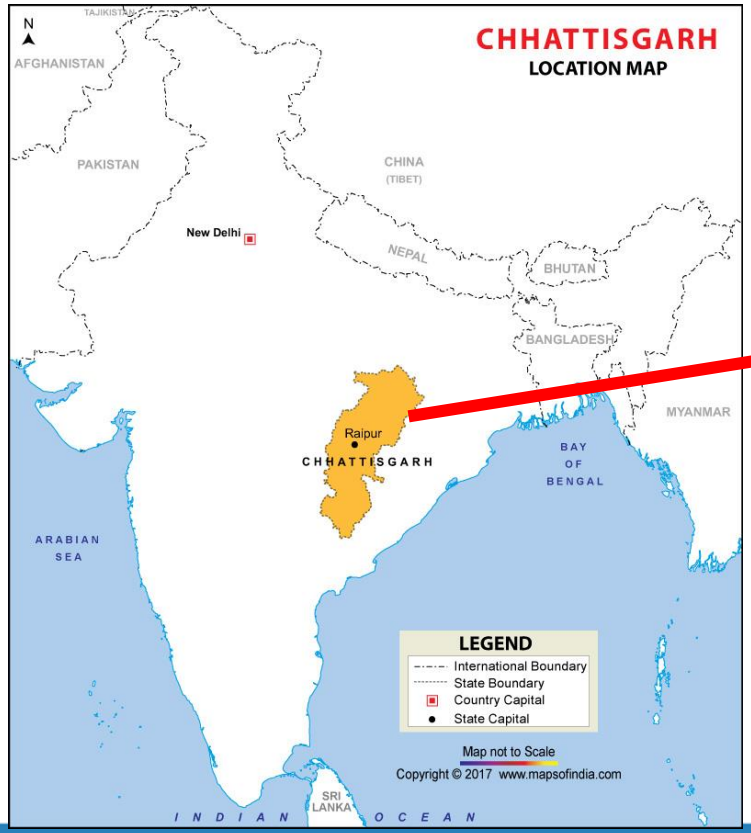
Transit oriented development plan for Naya Raipur:

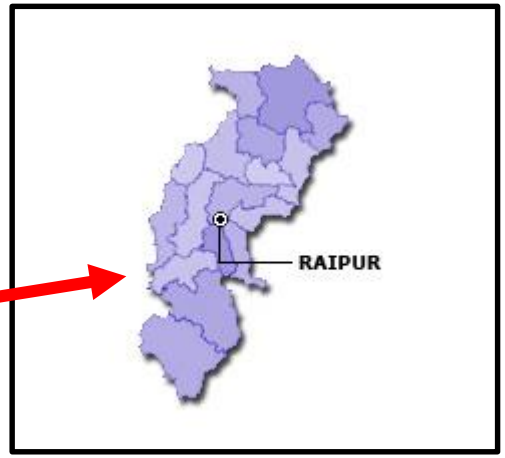
- Location of Naya Raipur
- Objectives of Naya Raipur
- Site selection of Naya Raipur
- Land use pattern
- Travel demand
- TOD model
- Outcomes & commercial land use
- Conclusion

Study area: Naya Raipur, Chhattisgarh

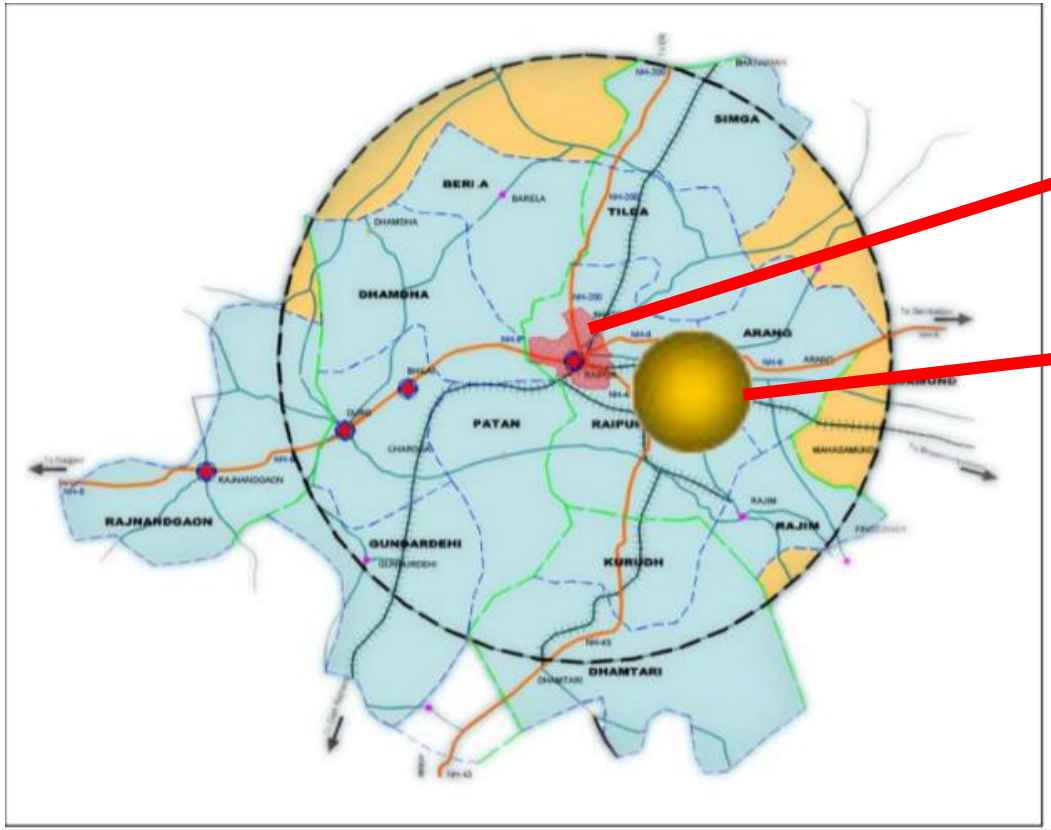
- Raipur is the capital city of Chhattisgarh state in India.
- A new capital city is planned named as ‘Naya Raipur’.
- It is a greenfield development project near Raipur.
- It is later renamed as ‘Naya Raipur-Atal Nagar’ honoring Ex Prime-minister of India.

Raipur, Chhattisgarh on map





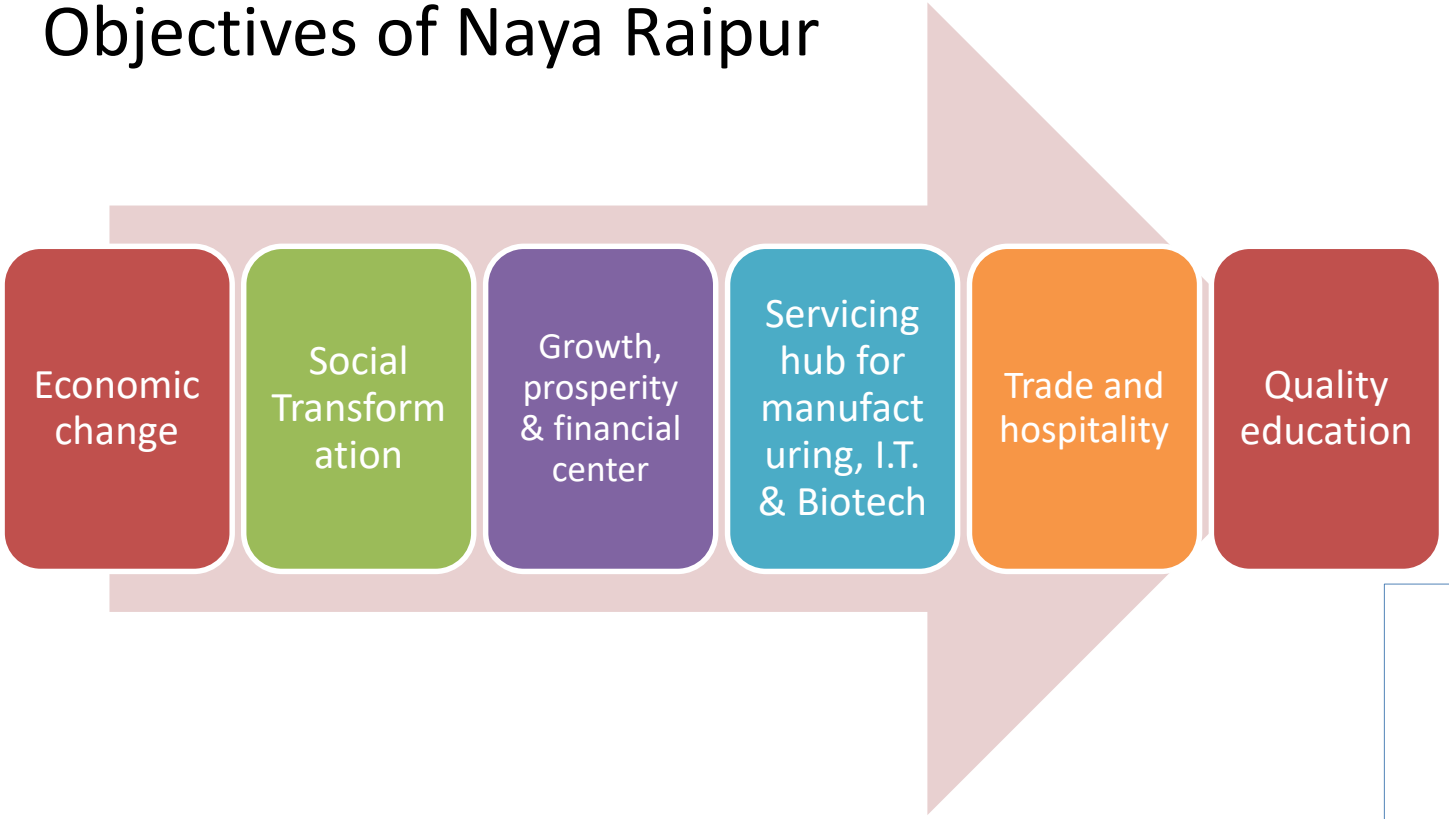
Study area: Naya Raipur, Chhattisgarh



Raipur

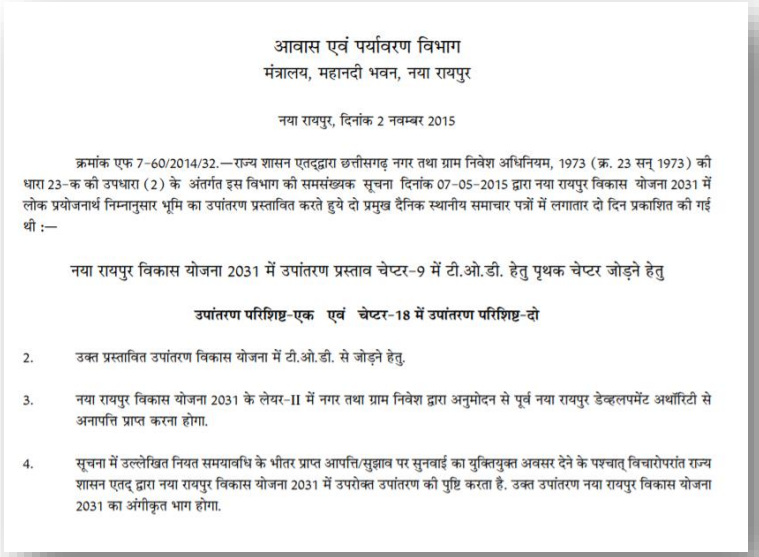
Naya Raipur

Objectives of Naya Raipur



TOD in Naya Raipur

- TOD was not formally inducted in master plan 2031 released in 2008.
- Later a special notification was issued containing changes in master plan in 2015.
- Notification introduced TOD in master plan, key definitions related to it and necessary changes for TOD planning.



आवास एवं पर्यावरण विभाग
मंत्रालय, महानदी भवन, नया रायपुर

नया रायपुर, दिनांक 2 नवम्बर 2015

क्रमांक एफ 7-60/2014/32.—राज्य शासन एतद्द्वारा छत्तीसगढ़ नगर तथा ग्राम निवेश अधिनियम, 1973 (क्र. 23 सन् 1973) की धारा 23-क की उपधारा (2) के अंतर्गत इस विभाग की समसंख्यक सूचना दिनांक 07-05-2015 द्वारा नया रायपुर विकास योजना 2031 में लोक प्रयोजनार्थ निम्नानुसार भूमि का उपांतरण प्रस्तावित करते हुये दो प्रमुख दैनिक स्थानीय समाचार पत्रों में लगातार दो दिन प्रकाशित की गई थी :—

नया रायपुर विकास योजना 2031 में उपांतरण प्रस्ताव चेप्टर-9 में टी.ओ.डी. हेतु पृथक चेप्टर जोड़ने हेतु

उपांतरण परिशिष्ट-एक एवं चेप्टर-18 में उपांतरण परिशिष्ट-दो

2. उक्त प्रस्तावित उपांतरण विकास योजना में टी.ओ.डी. से जोड़ने हेतु.

3. नया रायपुर विकास योजना 2031 के लेयर-II में नगर तथा ग्राम निवेश द्वारा अनुमोदन से पूर्व नया रायपुर डेव्हलपमेंट अथॉरिटी से अनापत्ति प्राप्त करना होगा.

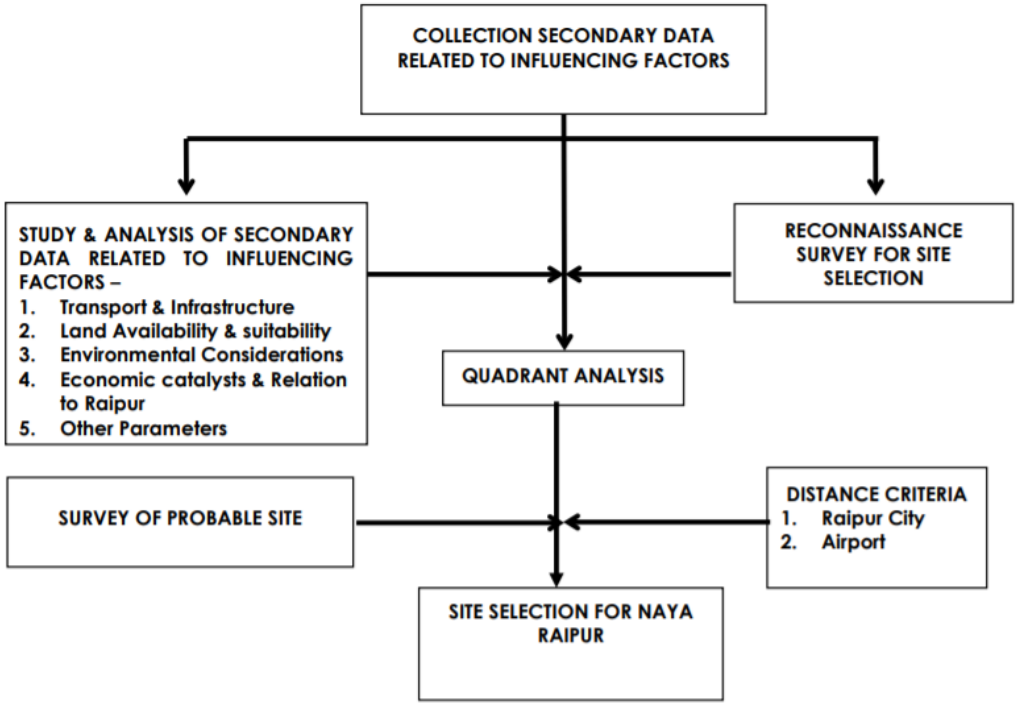
4. सूचना में उल्लेखित नियत समयावधि के भीतर प्राप्त आपत्ति/सुझाव पर सुनवाई का युक्तियुक्त अवसर देने के पश्चात् विचारोपरांत राज्य शासन एतद् द्वारा नया रायपुर विकास योजना 2031 में उपरोक्त उपांतरण की पुष्टि करता है. उक्त उपांतरण नया रायपुर विकास योजना 2031 का अंगीकृत भाग होगा.

Image: First page of notification released in 2015 for inducting TOD in master plan for Naya Raipur

Changes for TOD in master plan

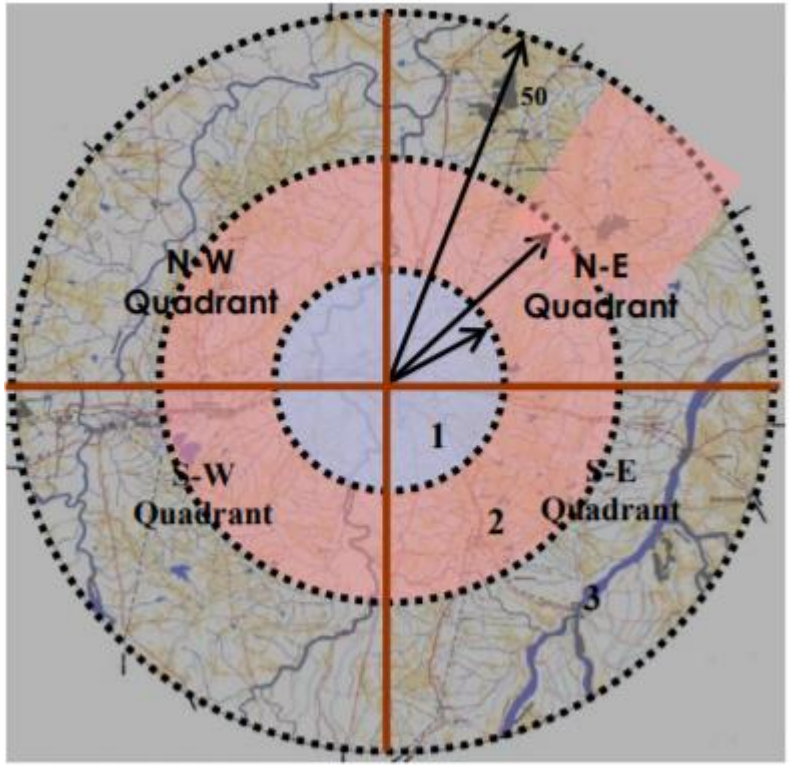
- Development code has been modified to permit the TOD.
- The main changes are related to:
  - Higher Density/compact development
  - Higher heights
  - Mixed land use
  - Street facing buildings
  - NMT movement

Site selection for Naya Raipur



Reconnaissance survey:
Survey carried out through visual inspection by expert

Quadrant analysis for site selection



Nearby area (radial distance-as per requirement) is divided into four parts:

- North-East
- South-East
- North-West
- South-West

and features of every quadrant are analyzed.

Radial distance of 50 KM is taken for quadrant analysis in case of Naya Raipur from Raipur as the centre.

Features of S-E Quadrant 1/2

- Availability of water due to proximity to Mahanadi River and Mahanadi Canal
- Absence of mining area
- Availability of barren/non-agricultural land
- Abundant availability of government owned land
- Proximity to Raipur City

After an analysis of the different quadrants based on the various parameters, southeast quadrant was found to be most suitable for the locating the proposed new Capital City

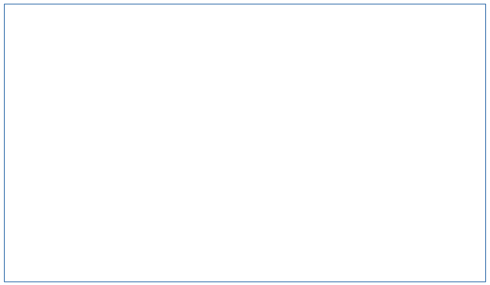
Features of S-E Quadrant 2/2

- Conducive traffic condition on NH-43 and NH-6 for providing accessibility to the new City
- Proximity to the Raipur-Vizainagaram and Mumbai-Kolkata Railway Line
- Presence of natural features like water bodies for recreational spots in the city
- Proximity to the Airport

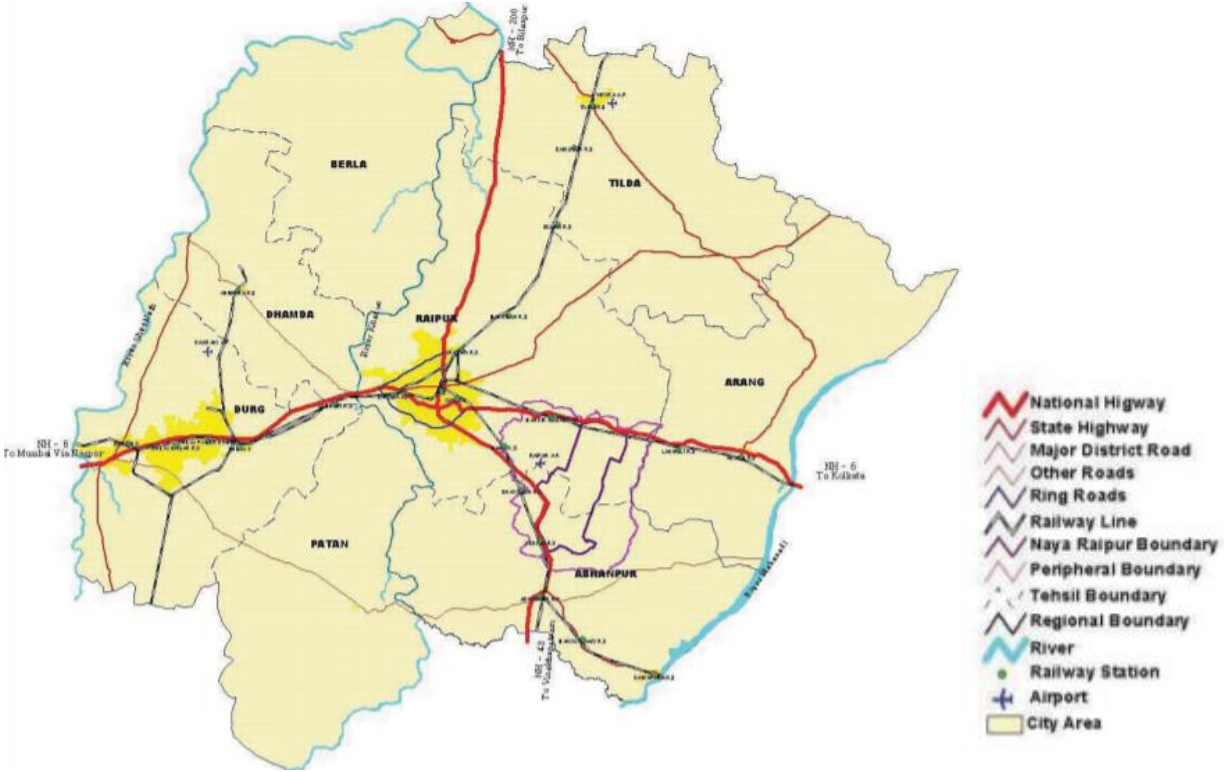
Site selection criteria: Transport & infrastructure

Existing linkage to

- Transport network and easy future connectivity
- Infrastructure network including
  - Water
  - Power
  - Telecommunications



Existing Transport infrastructure



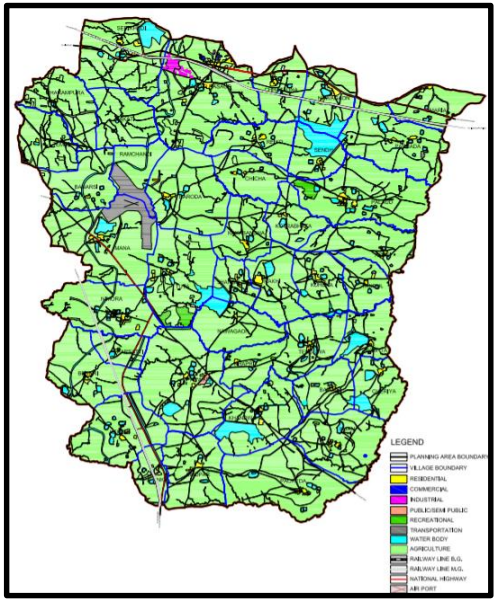
- There are four major roads within the site selection region.
- Three National Highways, namely; NH-6, NH-200 and NH-43 and one State Highway SH-5

Land availability & suitability

- Availability of Government land
- Land otherwise unsuitable for agriculture, mining
- Land with development friendly contours
- Low existing human settlements
- Minimum forest cover and wildlife
- Gentle slope to facilitate easy drainage and discharge of effluents
- Soil conditions with good bearing capacity for structural stability

Land use distribution: Existing

Land Use	Area (in Ha)	Percentage
Residential	257.48	1.08
Commercial	6.50	0.04
Industrial	29.07	0.12
Public & Semi Public (Public amenities and facilities)	20.59	0.08
Transportation - Roads and Streets	1131.44	4.77
Recreational	107.73	0.45
Water Body	1846.70	7.78
Agriculture	20343.12	85.68
Total	**23742.63**	**100**



- Most of the area is agricultural land.
- Very low population density
- Low commercial/recreational land use

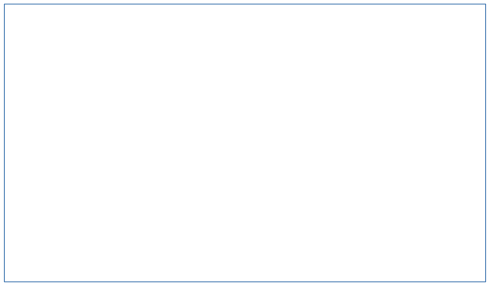
Transport Planning: Objective & integration

Transport plan is proposed with the objective of

- High mobility for all
- Providing mass transport system which is equitable and safe

The regional and city level transport is integrated linking the city with

- Regional rail
- Highway
- Airport



Travel Demand Assessment

- Naya Raipur is being planned for a population size of 5.6 lakh people.
- Travel demand on the transport network is assessed based on the adopted trip rates.
- The intra-city trip generation, by vehicular modes, is estimated with a per capita Trip Rate (PCTR) of 1.0 (including walk trips it would be about 1.35 to 1.5).
- The inter-city trips moving on intra-city network system is assessed at 20% of the intra-city trips

Trip generation & mode share: Planned

Category	Intra-city person vehicular trips per day	Inter-city person vehicular trips per day on intra-city network
Trips	540,000	108,000
Modal Share		
Private Modes	216,000	54,000
Public Transport	324,000	54,000
Sub Modal Share		
Bus	162,000	21,600
LRTS	97,200	21,600
IPT	64,800	10,800

LRTS- Light Rail Transit System; IPT- Intermediate Public Transport

- Total trips per day = 6.48 lakh
- Trips per day using public transport = 3.78 lakh (58%)
- Bus Trips = 1.83 lakh (28%)
- If LRTS is not developed, Bus trips would be about 2.93 lakh (45%)

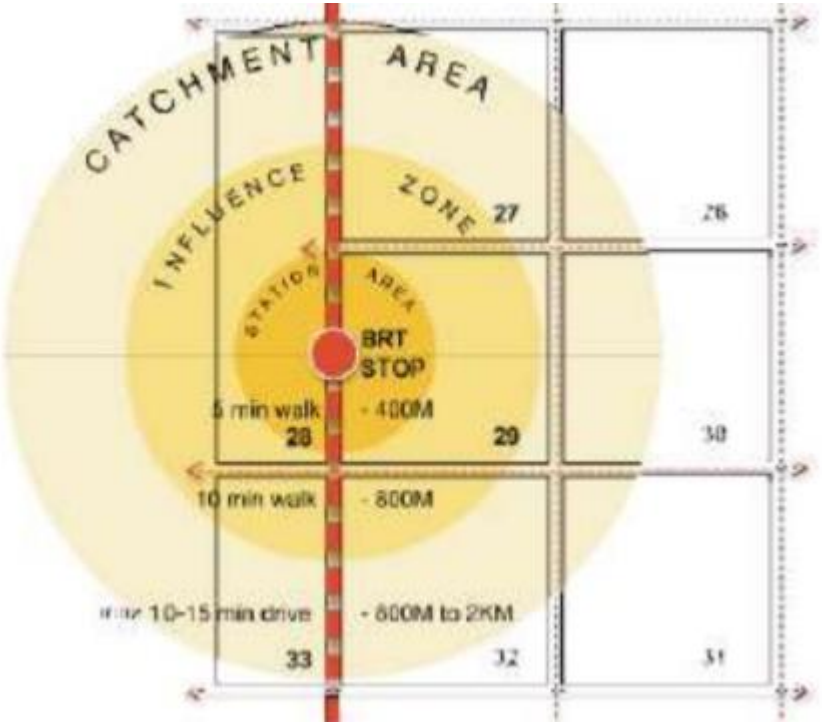
Average trip length would be about 4.5 km.

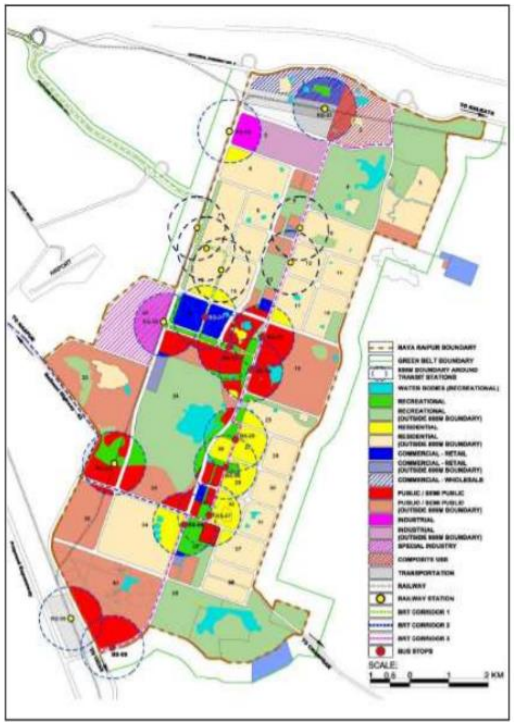
Importance of BRTS in Naya Raipur

Bus Rapid Transport System (BRT) would

- Form the core of a mass transportation plan for Naya Raipur.
- Dedicated bus lanes, cycling tracks and safe pedestrian walkways would complement the BRT
- Ease the development of an energy efficient and environmentally sustainable city.
- Supporting road sections and junctions would be designed to be friendly for children and persons with physical disabilities

TOD Zone in Naya Raipur

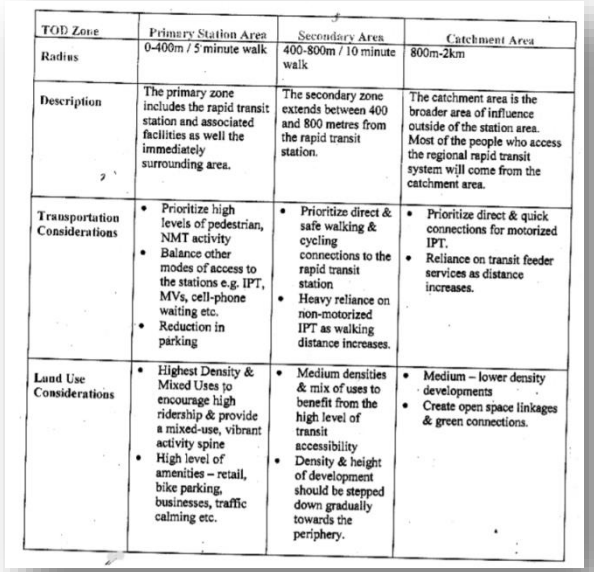




- Total Number Of 17 TOD zone planned.
- TOD zone is divided in 3 part
  - Station area
  - Influence zone
  - Catchment area

TOD zone: Station area

- 0-400m/5 min. walk
- Transit station and related facilities (ticket counter, shops etc.)
- Prioritized pedestrian area
- High density, mix use



TOD Zone	Primary Station Area	Secondary Area	Catchment Area
Radius	0-400m / 5 minute walk	400-800m / 10 minute walk	800m-2km
Description	The primary zone includes the rapid transit station and associated facilities as well the immediately surrounding area.	The secondary zone extends between 400 and 800 metres from the rapid transit station.	The catchment area is the broader area of influence outside of the station area. Most of the people who access the regional rapid transit system will come from the catchment area.
Transportation Considerations	• Prioritize high levels of pedestrian, NMT activity • Balance other modes of access to the stations e.g. IPT, MVs, cell-phone waiting etc. • Reduction in parking	• Prioritize direct & safe walking & cycling connections to the rapid transit station • Heavy reliance on non-motorized IPT as walking distance increases.	• Prioritize direct & quick connections for motorized IPT. • Reliance on transit feeder services as distance increases.
Land Use Considerations	• Highest Density & Mixed Uses to encourage high ridership & provide a mixed-use, vibrant activity spine • High level of amenities – retail, bike parking, businesses, traffic calming etc.	• Medium densities & mix of uses to benefit from the high level of transit accessibility • Density & height of development should be stepped down gradually towards the periphery.	• Medium – lower density developments • Create open space linkages & green connections.

Image: Notification describing TOD zone

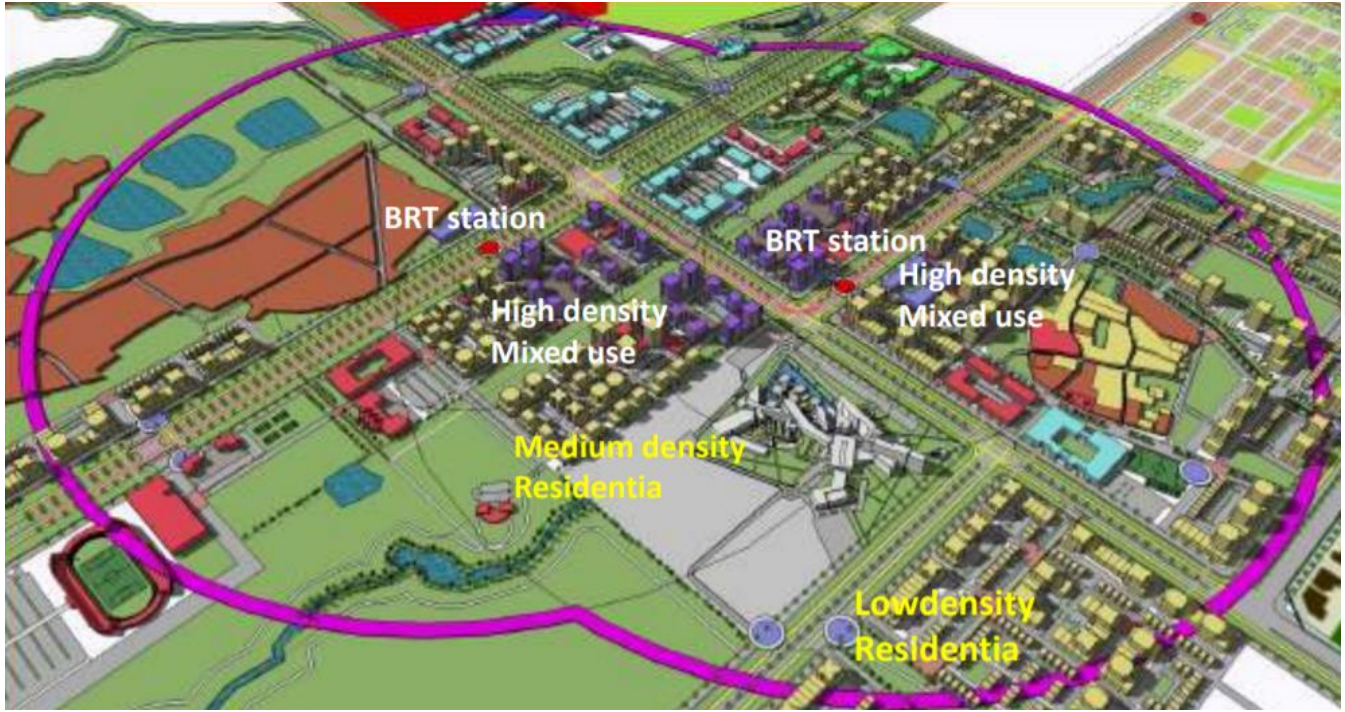
TOD zone: Influence zone

- 400-800 meters or 10 min. walk area
- High feasibility for cycling and walking
- Medium densities and mix land uses
- Descending height of building towards periphery

TOD zone: Catchment area

- 800 meter to 2 km. area
- Facilitating both NMT and motorized vehicles.
- Feeder buses or similar facility provided to attract trips for transit
- Medium to low density development
- Open spaces and green path (exclusive for cycle & pedestrian provided)

Model depiction of TOD zone



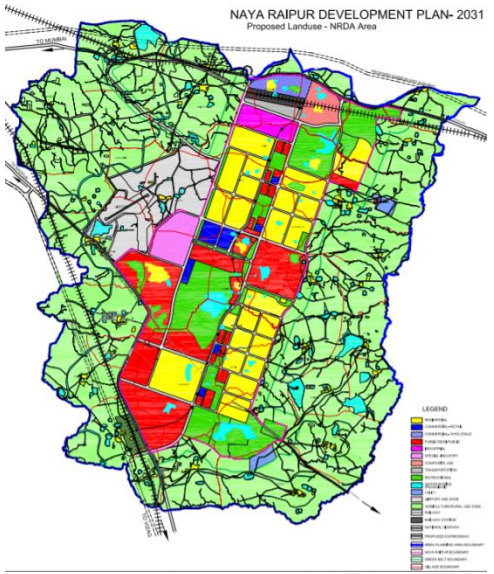
As distance from node/station increases,

- Denseness reduces
- Mix use reduces
- High rise development reduces

Land use distribution: Planned

S.No.	Land Use	Area (ha)	Percentage
1	Residential	2113.39	26.37
2	Commercial - Retail	146.67	1.81
3	Commercial - Wholesale	130.67	1.63
4	Industrial	196.13	2.42
5	Special Industry	263.05	3.28
6	Public & Semi Public	1846.38	23.04
7	Recreational	2137.44	26.67
8	Transport	1005.77	12.55
9	Composite Use	177.6	2.22
	Total	**8013.1**	**100**

- High population density.
- Land use for offices, transport and recreation increased drastically.

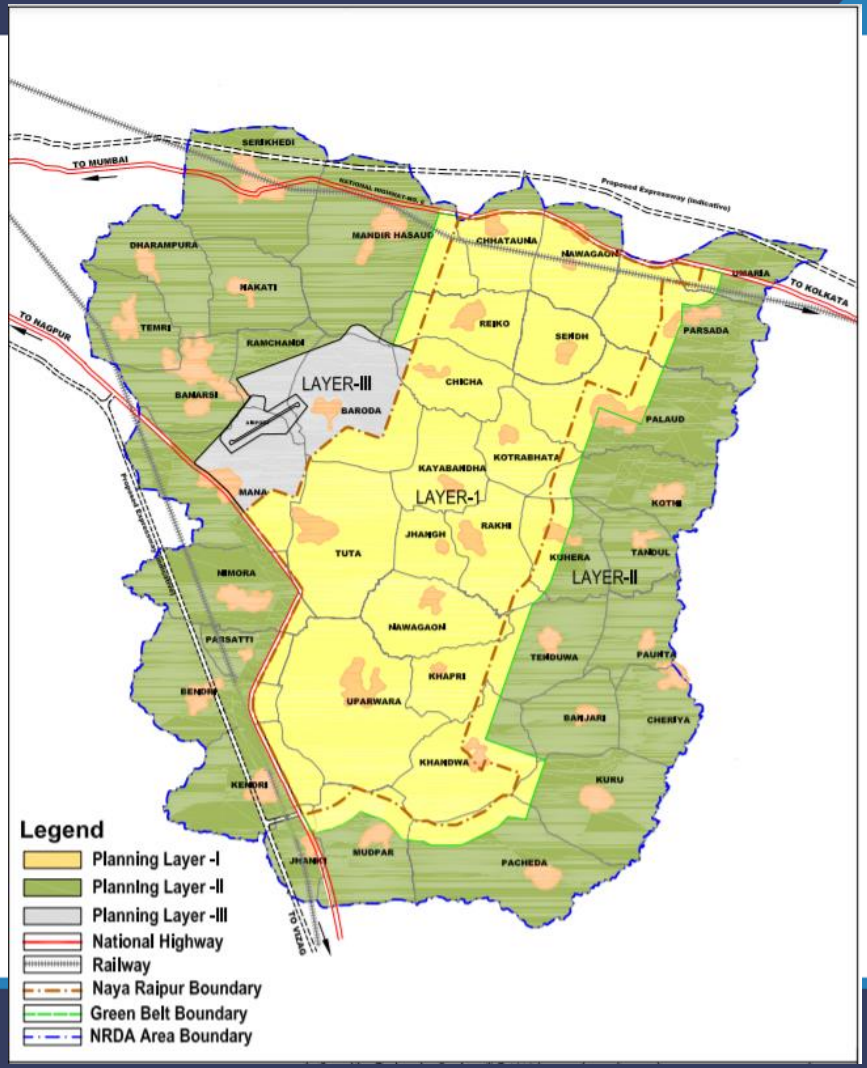


Division of Site: as per land use

- NRDA (New Raipur Development Authority) area is divided into 3 layers.
- Delineating the Naya Raipur, the peripheral and the airport zone.

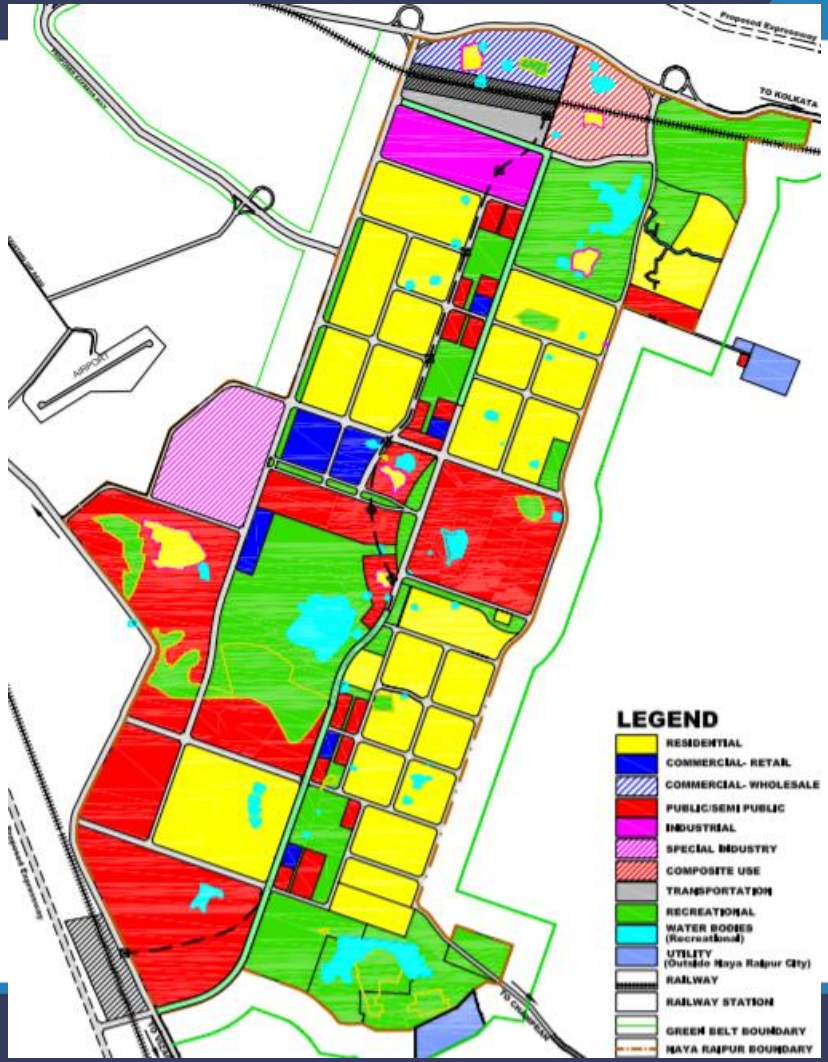
Layer	Area Included
Layer-I	The Proposed Naya Raipur (Area-80.13 sq.km.) Including 500 – meter wide green belt (Area-95.22 sq. km.)
Layer-II	Naya Raipur Peripheral Region (NRPR) (Area-130.28 sq.km.)
Layer-III	Airport Zone (Area-11.92 sq.km.)

Division of Site: as per land use



- Layer-1 is core area with dense development containing all important offices, commercial centers and housing.
- Layer-2 is a green area with continued land use as agriculture & forest.
- Layer-3 is airport zone

Land use pattern: Planned in layer-I



TOD zone example: Sector-31





Recreational spots around water bodies

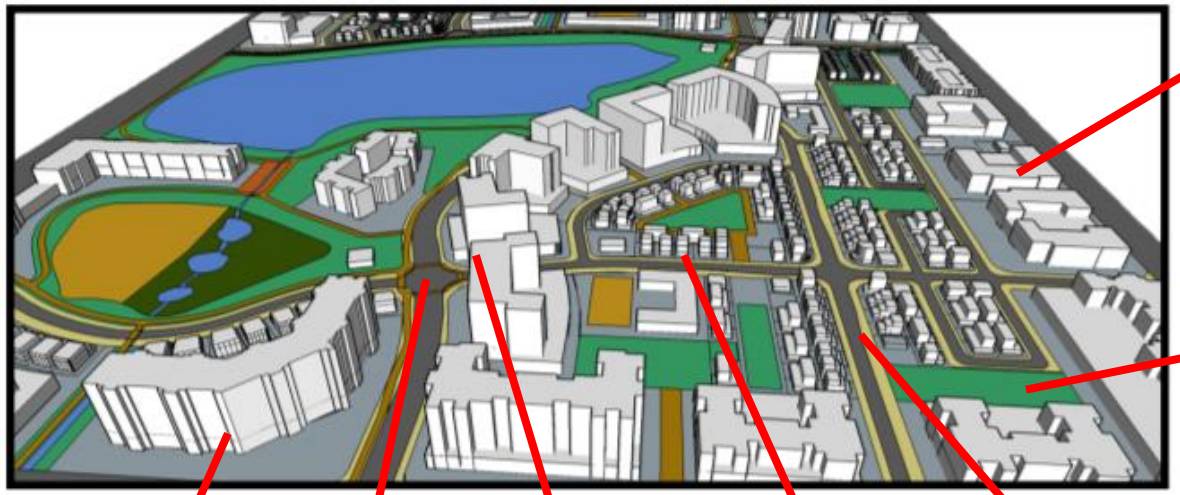
Transit

Road facing high rise buildings

Open spaces

Connecting roads

TOD plan for Sector 31



Residential buildings within walking distance to transit

Open space/park

Mall/commercial center

Transit

High rise office buildings facing roads

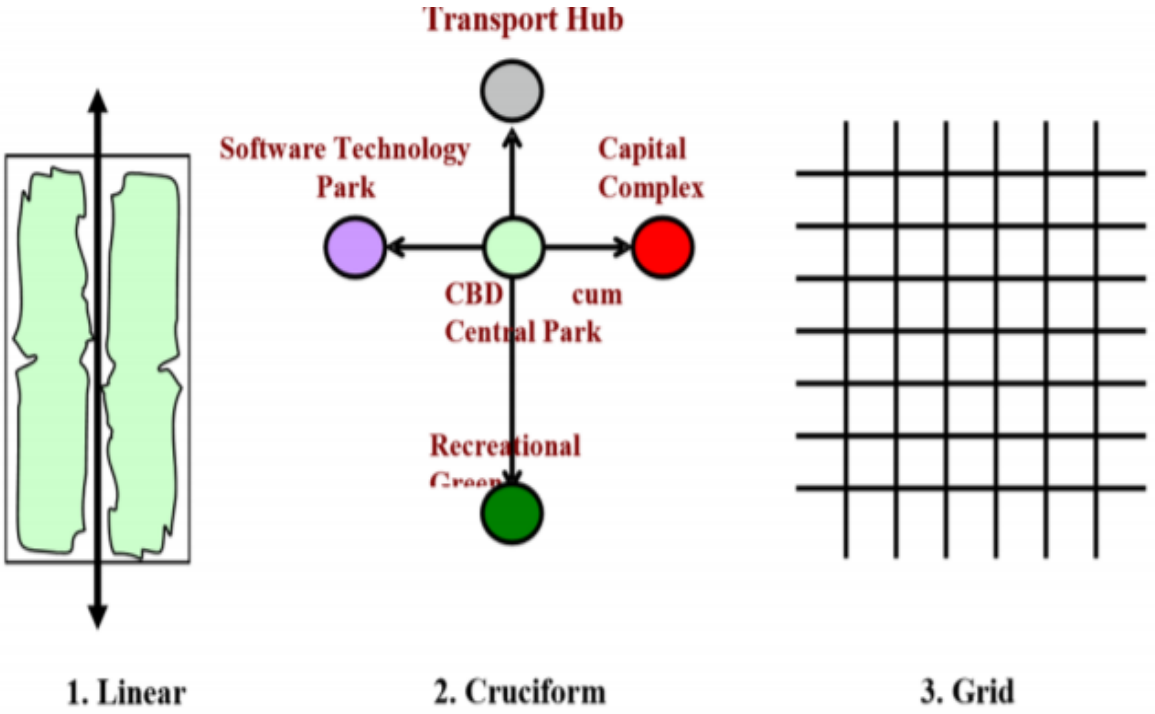
Medium rise buildings

Supporting transport infra.

Positive outcomes of TOD planning

- 100 % population within walking distance/TOD zone
- More pedestrian Friendly and sustainable environment
- Multiple choice of Travel Modes
- Safe, Attractive and vibrant living environment
- Dense and compact development
- More financial returns thus easy to get funding

Development pattern in Naya Raipur



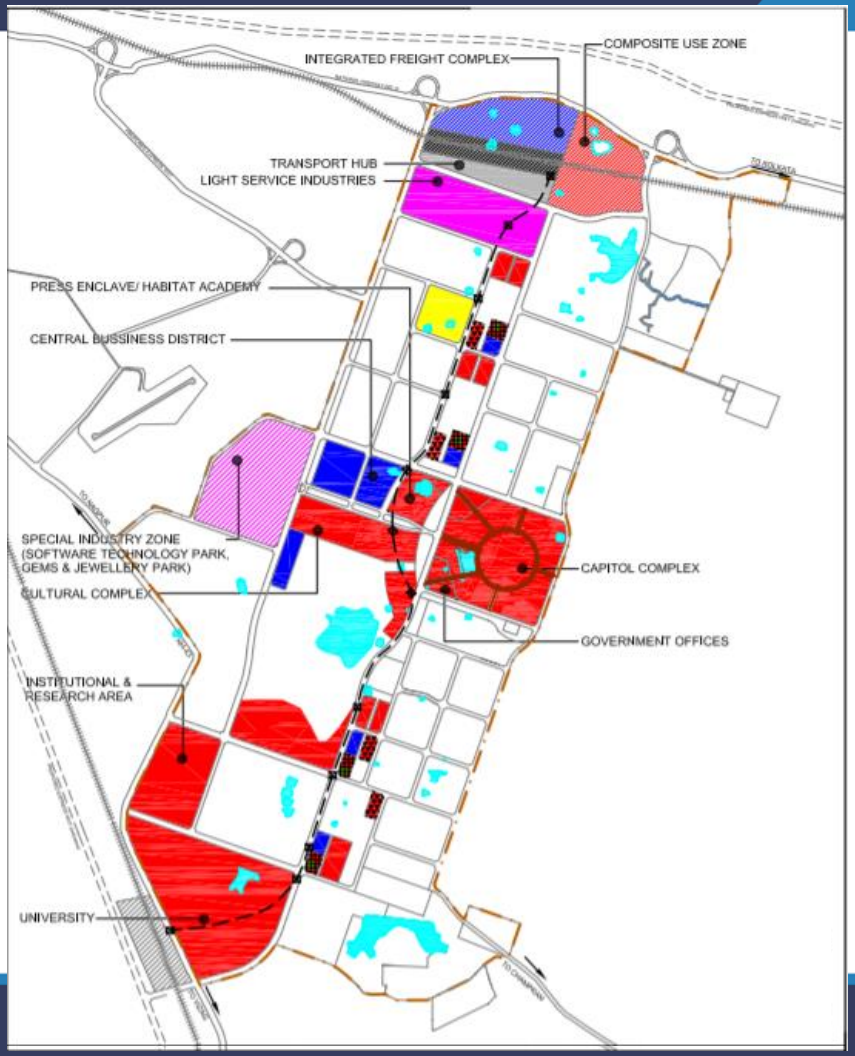
Naya Raipur development will follow linear, cruciform and grid pattern.

Transport network will also follow grid pattern with BRTS transit as most broad road.

Facility	Population Served	Area* (Ha)	Activities
City Centre/CBD	5 lakh	96.12 ha	Shopping (Retail Service, Repair & limited Wholesale) Informal Shopping, Commercial Offices, Cinema, Hotel, Guest House, and Nursing Home.
			Auditorium, Museum, Library, Science Centre, Art/Craft/Music/ Dance School, Craft/Mela/Book Bazar, Weekly Markets (on close days), Local Government Offices.
			Bus Terminal, Fire Post, Police Post, Telephone Exchange, Electric Sub Station, Post and Telegraph Office, Petrol Pump, Conveniences and Residential
Community Shopping & Business Complex	1.0 to 1.5 lakh	5.4 ha	Shopping (Retail Service, Repair, Informal Shopping, Commercial Offices, Cinema, Hotel, Guest House, Nursing Home)
			Post office, Dispensary, Petrol Pump (filling Station only) Weekly Markets (on off days) Electric Sub-Station Conveniences
Sector / Neighbourhood Shopping Centre	15,000 to 20,000	0.46 ha	Shopping Retail Service, Repair, Informal Shops, and Commercial Offices.
			Community Hall and Library
			Electric Sub-Station Conveniences
Housing Area Centre	About 5 thousand	0.11 ha	Shopping Retail Service, Repair, Informal Shopping
			Electric Sub Station Conveniences.

Commercial activity along transit

Commercial land use along transit



Housing pattern along TOD: Planned

S. No.	Housing Sub-system	Number of Dwelling Units	Percentage of Housing stock (Naya Raipur)
1	Private Housing	74,602	60.23
2	Co-operative group housing societies	14,430	11.65
3	Housing for Government Employees	6,315	5.1
4	VIP Housing	2,666	2.15
5	Private Employee Housing	2,220	1.79
6	Institutional Housing	7,780	6.28
7	Others (CBD, Composite use, Facility corridor)	9,287	7.5
8	Urban Villages	6,560	5.3
	Total	**123,860**	**100**

Housing accommodating various sections of society from low-income group to VIP housing.

Images of Naya Raipur: Roundabouts for heavy traffic



Roads and roundabouts are designed for high flow of traffic with necessary safety measures.

Image: yourstory.com

Images of Naya Raipur: Housing units



Image: patrika.com

Inclusive housing is planned.

Housing for low-, middle- and high-income group.

Images of Naya Raipur: High rise development



High rise, medium rise and low-rise development according to planning in Naya Raipur.

Image: indiatoday.in

Conclusions

- Naya Raipur is a greenfield city developed with BRTS system.
- A feasible site is selected.
- TOD principles such as high-rise, compact, mix development along transit are applied during development.
- TOD implementation will help city to generate financial profits, jobs and high living standard city for residents.

References

- Naya Raipur Development Authority (NRDA); 2008 “Naya Raipur development plan 2031”

- Government of Chhattisgarh; December 4th, 2015 “Gazette notification for addition of TOD in Naya Raipur master plan 2031”

Thank You